

Sandra Sudevan
192424422

SET-4

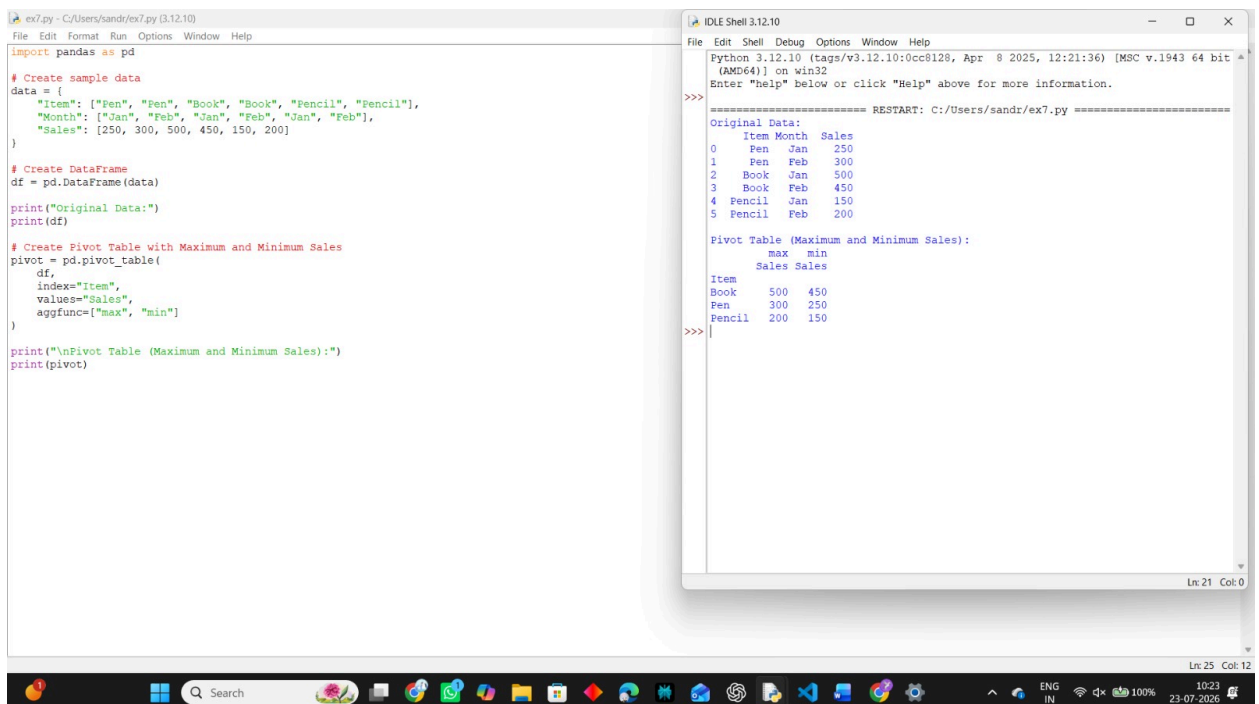
7. Write a Pandas program to create a Pivot table and find the maximum and minimum sale value of the items.

Aim:

To create a Pivot table and find the maximum and minimum sale value of items.

Procedure:

1. Import the Pandas library.
2. Create the sales dataset.
3. Use the `pivot_table()` function.
4. Apply `max` and `min` aggregation functions.
5. Display the Pivot table.



```
ex7.py - C:/Users/sandr/ex7.py (3.12.10)
File Edit Format Run Options Window Help
import pandas as pd

# Create sample data
data = {
    "Item": ["Pen", "Pen", "Book", "Book", "Pencil", "Pencil"],
    "Month": ["Jan", "Feb", "Jan", "Feb", "Jan", "Feb"],
    "Sales": [250, 300, 500, 450, 150, 200]
}

# Create DataFrame
df = pd.DataFrame(data)
print("Original Data:")
print(df)

# Create Pivot Table with Maximum and Minimum Sales
pivot = pd.pivot_table(
    df,
    index="Item",
    values="Sales",
    aggfunc=["max", "min"]
)

print("\nPivot Table (Maximum and Minimum Sales):")
print(pivot)
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/ex7.py =====
Original Data:
  Item Month Sales
0  Pen   Jan   250
1  Pen   Feb   300
2  Book   Jan   500
3  Book   Feb   450
4 Pencil   Jan   150
5 Pencil   Feb   200

Pivot Table (Maximum and Minimum Sales):
      max min
Item  Sales Sales
Book    500 450
Pen     300 250
Pencil  200 150

>>>
```

Result:

The maximum and minimum sale values of each item were displayed successfully.

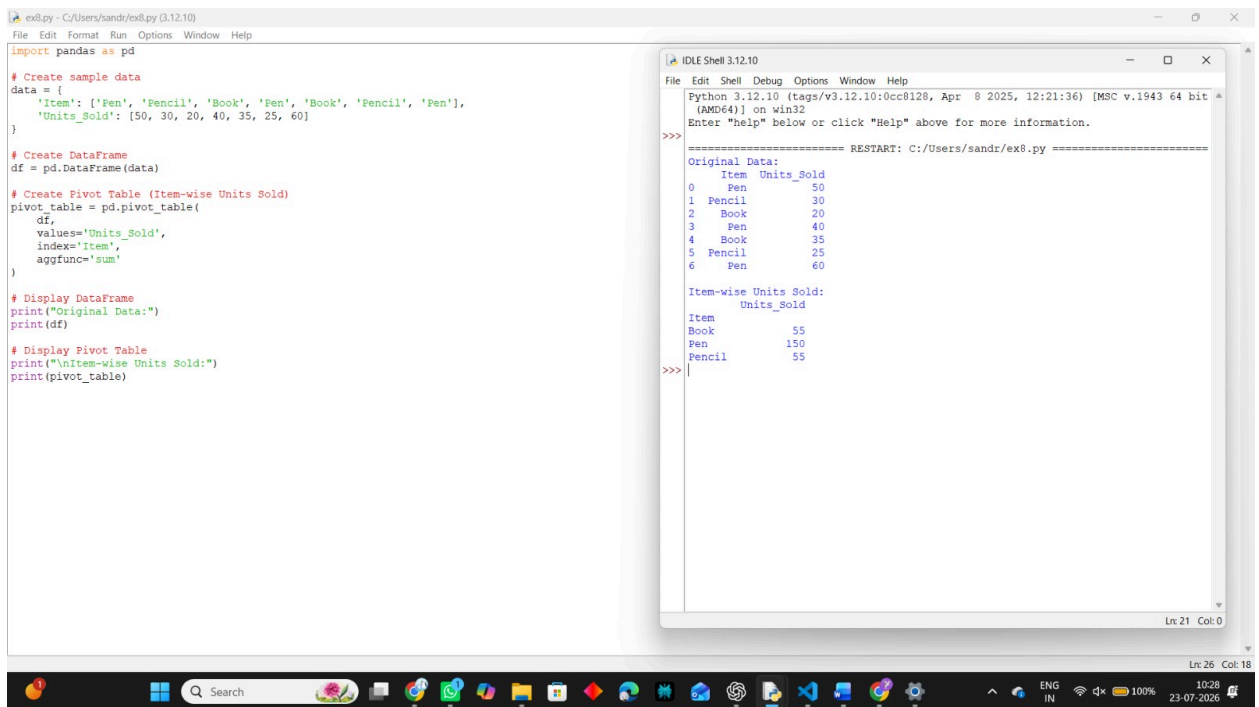
8. Write a Pandas program to create a Pivot table and find the item wise unit sold.

Aim:

To create a Pivot table and find the item-wise units sold.

Procedure:

1. Import the Pandas library.
2. Create the sales dataset.
3. Use the `pivot_table()` function.
4. Apply the `sum` aggregation function on the Units Sold column.
5. Display the Pivot table.



The screenshot shows a Python IDE with two windows. The left window displays the source code for a program that creates a sample dataset, converts it to a DataFrame, and then uses the `pivot_table()` function to calculate the sum of units sold for each item. The right window shows the output of the program, displaying the original data and the resulting pivot table.

```
import pandas as pd

# Create sample data
data = {
    'Item': ['Pen', 'Pencil', 'Book', 'Pen', 'Book', 'Pencil', 'Pen'],
    'Units_Sold': [50, 30, 20, 40, 35, 25, 60]
}

# Create DataFrame
df = pd.DataFrame(data)

# Create Pivot Table (Item-wise Units Sold)
pivot_table = pd.pivot_table(
    df,
    values='Units_Sold',
    index='Item',
    aggfunc='sum'
)

# Display DataFrame
print("Original Data:")
print(df)

# Display Pivot Table
print("\nItem-wise Units Sold:")
print(pivot_table)
```

Original Data:

	Item	Units_Sold
0	Pen	50
1	Pencil	30
2	Book	20
3	Pen	40
4	Book	35
5	Pencil	25
6	Pen	60

Item-wise Units Sold:

Item	Units_Sold
Book	55
Pen	150
Pencil	55

Result:

The item-wise units sold were displayed successfully.